PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : THERMODYANAMIC & THERMOCHEMISTRY (Question Bank # 03)

SINGLE CORRECT

1. **(C)**

Sol. $\Delta H = \frac{5}{2} \times 200 = 500 \text{ kJ per mol B}$

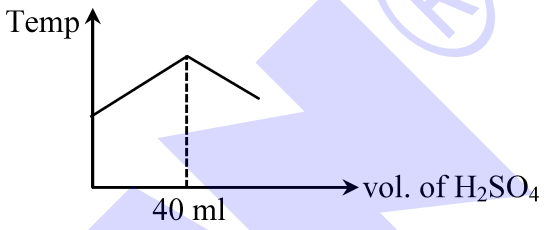
2. **(B)**

Sol. $N_1V_1 = N_2V_2$

$0.4 \times 50 = 0.25 \times 2 \times V_2$

$V_2 = 40 \text{ cm}^3$

$\Delta t = \frac{q}{ms}$



3. **(A)**

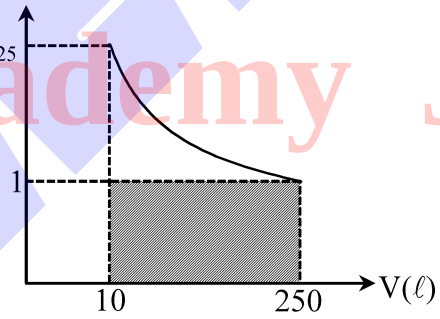
Sol. $P_1V_1 = P_2V_2$

$25 \times 10 = P_{250} = 1 \times V_2$

$V_2 = 250\ \ell$

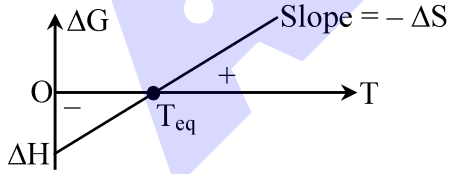
$w = -P_{ext}(V_2 - V_1)$

$= -1(250 - 10) = -240\ \ell\text{-atm}$



4. **(B)**

Sol. If $\Delta H < 0$ & $\Delta S < 0$ then spont only at low temperature



5. **(D)**

Sol. In adiabatic free expension of ideal gas its temp. remains constant, while in adiabatic reve & adiabatic single stage expansion temperature decreases. Hence $T_B < T_C < T_R = T_A$

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6. **(A)**

Sol. Amount independent ⇒ Intensive

Path independ ⇒ state function

U	Sp. enthalpy = $\frac{H}{mass}$	$q_p = \Delta H$	$q_r = \Delta U_m$	T	w
Intensive ✗	✓	✗	✓	✓	✗
Path independ ✓	✓	✓	✓	✓	✗

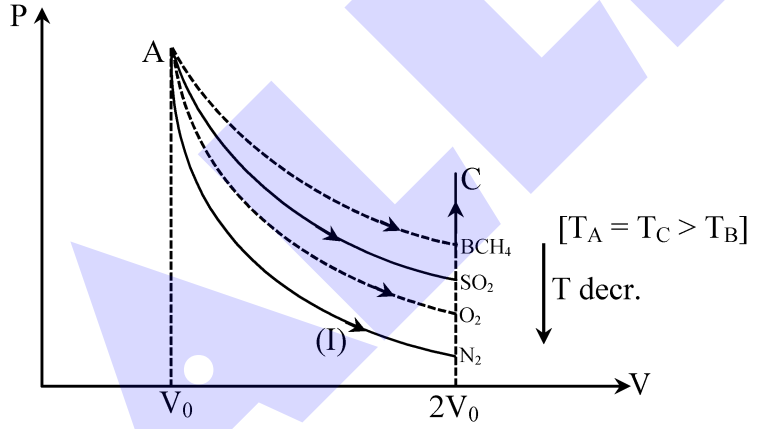
7. **(A,C)**

Sol.

	T_f	R_f	$V_f = 3N - (T_f + R_f)$	C_v	C_p	r
Ne	3	–	–	$\frac{3}{2}R$	$\frac{5}{2}R$	1.67
O_2	3	2	1	$\frac{7}{2}R$	$\frac{9}{2}R$	1.29
SO_2	3	3	3	6R	7R	1.17
CH_4	3	3	9	12R	13R	1.08

$\left\{\left|\frac{dP}{dV}\right| \propto \gamma \text{ for Reversible Adiabatic}\right\}$

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ΔU_{AB}	+	ΔU_{BC}	=	ΔU_{AC}
↓		↓		↓ = $nC_V\Delta T$
$q_{AB} + w_{AB}$ = 0		$q_{BC} + w_{BC}$ = 0		0 (Temp. Same)
Adiab.		(Isochoric)		

$\boxed{q_{BC} = -w_{AB}}$ for CH_4 in step-I $-w_{AB}$ is max.

Hence for CH_4 in step II q_{BC} is max.

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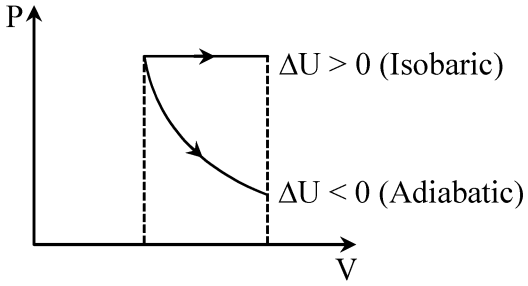
8. **(B)**

Sol. (A) $w = \Delta U$

(B) $w = \Delta U = nC_v\Delta T$

(C) Since temp. is equal K.E. is same

(D)



9. **(B,D)**

Sol. (A) $(\Delta_f H^\circ)_{H_2O(\ell)} = (\Delta_C H^\circ)_{H_2(g)} < 0$)

(B) $O_2(g) \rightarrow O_2(g)$ $\Delta_r S^\circ = \Delta_f S^\circ = 0$

(C) $S^\circ[O_2(s)] = 0$ at 0 K

(D) $(S^\circ)_{H^+(aq)}$ has been assigned zero hence ions may show + or – entropy compared to H^+.

10. **(B,C,D)**

Sol. (A) $C(gr, S) \rightarrow C(gas)]\ \Delta H_{atom} = \Delta_f H_{c(gas)}$

(B) $H_2(g) + \frac{1}{2}O_2 \rightarrow H_2O_{(\ell)}]\ (\Delta_f H)_{H_2O(\ell)} = (\Delta_f H^0)_{H_2(g)}$

(C) $(\Delta_f H^\circ)_{H_2O(\ell)} < 0$

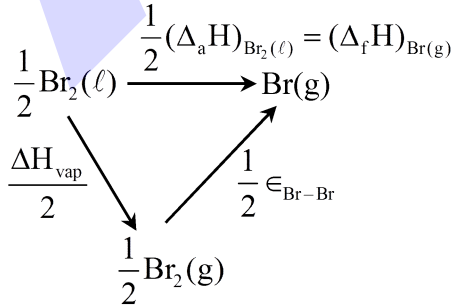
(D) $C(gr, s) + O_2(g) \rightarrow CO_2(g)]\ (\Delta_C H)_{C(qr, s)} = (\Delta_f H^\circ)_{CO_2(g)}$

11. **(A,B)**

Sol. (A) $\frac{1}{2}H_2(g) \rightarrow H(g)]\ (\Delta_f H)_{H(g)} = \frac{1}{2}(\Delta aH)_{H_2} = \frac{1}{2}\varepsilon_{H-H}$

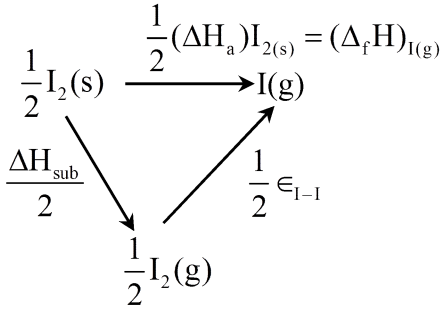
(B) $C(\text{solid, graphite}) \rightarrow C(gas)]\ (\Delta_f H)_{C(gas)} = (\Delta aH)_{C(qr, s)} = (\Delta H_{sub})_{C(qr,s)}$

(C)



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(D)



12. (C)

Sol. $N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$ may be obtained by

$$b\left(-\frac{1}{4}\right)+c\left(\frac{1}{4}\right)+a\left(-\frac{3}{4}\right)+d\left(-\frac{9}{4}\right)$$

$\because$ $\Delta H = -60 + 9 + 57 - 153 = -147 \frac{K_{cal}}{mol}$

13. (C)

Sol. $\Delta_r H^\circ = \Sigma(\Delta_f H^\circ)_{Prod} - \Sigma(\Delta_f H^\circ)_{React}$

$-147 = [0 + 2(-68)] - [(\Delta_f H^\circ)_{N_2H_4} + 0]$

$(\Delta_f H^\circ)_{N_2H_4} = 11$

14. (C)

Sol. $\Delta H_b > \Delta H_c$

INTEGER

1. (99.50)

Sol. $\Delta H = \Delta U + P_2V_2 - P_1V_1$

q + w

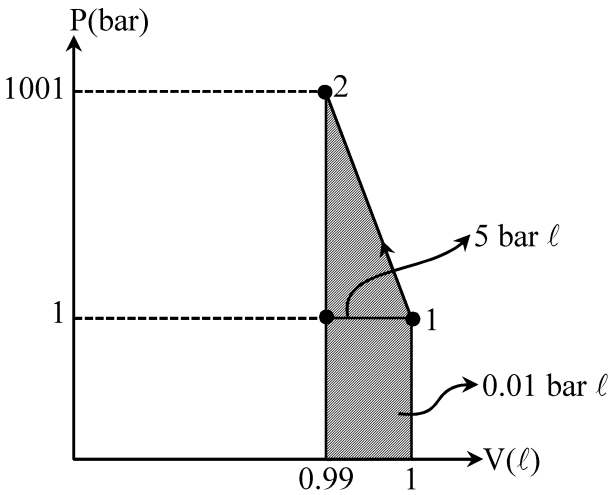
0 + Area

(Adiabatic)

x = Area + $[1001 \times 0.99 - 1 \times 1] \times 0.1$

5.01×0.1

$= 99.50$ kJ



1 bar. ℓ = 100 J = 0.1 kJ

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2. **(–2.40)**

Sol. $w = -\text{Area enclosed}$

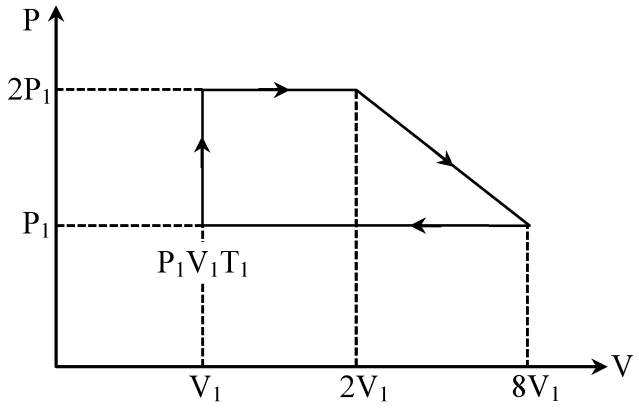
$= -[P_1V_1 + 3P_1V_1]$

$= -4P_1V_1$

$= -4nRT_1$

$= -4 \times 1 \times \frac{2}{1000} \times 300$

$= -2.40$ kcal



3. **(–17.80)**

Sol.

	$2A(s) \rightleftharpoons$	$B(g)$	+	$2C(g)$	+	$3D(g)$
PPr at eqm. (bar)	—	$12 \times \frac{1}{6} = 2$		$12 \times \frac{2}{6} = 4$		$12 \times \frac{3}{6} = 6$

$K_P^o = P_B(P_C)^2(P_D)^3 = 2(4)^2(6)^3 = 2^8 \times 3^3$

$\Delta_r G^o = -RT \ln k_p^o$

$= -\frac{2}{1000} \times (727 + 273)\ [8\ln 2 + 3\ln 3]$

$= -2 \times 8.9 = -17.8$ Kcal

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MATCH THE COLUMN

1. **(A) - P,R,S,T; (B) - P,T; (C) - P,R,S,T; (D) Q**

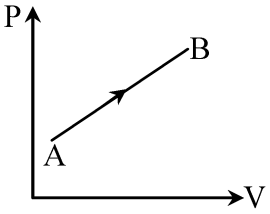
Sol. (A) $q > 0$ $[\Delta U = q + w]$

$+$? (Expansion)

$\Delta U > 0$ $(T\uparrow)$

$\Delta H > 0$ $(T\uparrow)$

Answer : P,R,S,T



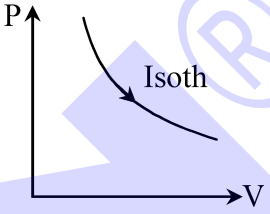
(B) $\Delta U = q + w$

$0 \quad + \quad -$

(expansion)

$\Delta H = \Delta U = 0$ (Isoth)

Answer : P, T

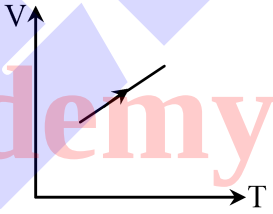


(C) $\Delta U = q + w$ $\Rightarrow q > 0$

$+ \quad -$ $\Delta U > 0$

$(T\uparrow)$? (Exp) $\Delta H > 0$

Answer : P,R,S,T



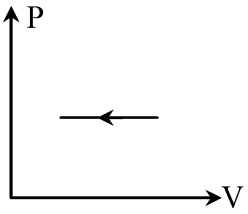
(D) $q_p = nC_p\Delta T < 0$ $(\because \Delta T < 0)$

$w > 0$ (compression)

$\Delta U < 0$

$\Delta H < 0$

Answer : Q



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